

# Daniel Lu

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## Education

### Carleton University, B.Sc. in Computer Science (Honours, Co-op)

Ottawa, Canada

AI & Machine Learning Stream - GPA: 3.9/4.0

Sep 2023 – Apr 2028

- **Coursework:** Supervised/Unsupervised Methods, Transformer Architectures, Language Modeling & Embeddings, Quantum Computing

## Experience

### Software Developer

May 2026 – Aug 2026

Shopify

Toronto, Canada

- Cut database load by optimizing or eliminating **40%** of inefficient read/write queries across **Ruby** and **Rust** payment services, reducing p95 query latency by **35%** and freeing **20%** of capacity for peak-traffic checkout load.
- Scaled Shopify Payments throughput **2.1x**, from **20,000** to **42,000** checkout-starts/minute, by conducting **8+** staged production rollouts and building **Grafana** dashboards for operational excellence that cut incident detection time by **50%**.

### Software Developer

May 2025 – Aug 2025

TrendAI

Ottawa, Canada

- Pioneered TrendAI's **first MCP**-based digital twin prototype, architecting a **Neo4j** graph **RAG** pipeline for predictive infrastructure security analytics, directly securing C-suite approval for full-scale development.
- Improved distributed system latency by **30%** and reduced monthly cloud costs by **18%** across **120+** **active enterprise** users via **AWS** cloud infrastructure optimizations and **Kubernetes** Horizontal Pod Autoscaling.
- Engineered **CI/CD** pipelines for internal SDKs and database deployment systems serving **200+** **developers** and **10,000+** **stakeholders**, reducing release cycle duration by **65%** (**10 hours** → **under 4 hours**).

### Software Developer

Jan 2025 – Apr 2025

Blackberry QNX

Ottawa, Canada

- Reduced system validation cycle time by **19%** by engineering automated testing scripts in **Python**, **C**, and **Bash** for **QNX** **RTOS** build pipelines, cutting manual verification effort from **8 hours** to **6.5 hours** per release.

## Projects

### Beat Battle | [Website](#) | [GitHub](#)

- Co-founded and developed a competitive multiplayer game for real-time music production on a stateless **Bun** + **Elysia** backend, using **Redis** pub/sub to synchronize **2000+** **concurrent players** across **400+** **live lobbies** at **sub-100ms** latency.
- Reached **150k+** **accounts** and **1.2M+** **unique site visits** within 13 days of launch with zero marketing spend, driven entirely by organic word-of-mouth and a rapidly growing **50,000-member** Discord community.

### Mime | [GitHub](#)

- Bridged the visual-audio immersion gap in cross-lingual video calls by engineering a lightweight real-time 3D facial overlay that renders lip-synced translated speech with **91% accuracy**.
- Architected and trained a **CNN-Transformer** audio encoder in **PyTorch** on the 76-hour **BEAT** motion-capture dataset, regressing 52 **ARKit** facial blendshape coefficients at **0.0098 MSE**.
- Shipped a zero-client-modification pipeline compatible with all major video conferencing platforms, achieving **<250ms** end-to-end streaming latency and **100ms TTFT**.

### Tobio | [Website](#) | [GitHub](#)

- Built a sensor-free 3D player tracking system from standard game footage using a monocular 3D reconstruction pipeline, deploying the product to **3 university teams** + **Brazilian men's national team** with no specialized hardware.
- Cut post-game analysis time by **85%** by piping a sliding-window action classifier into a vector-indexed **RAG** pipeline, enabling natural language querying of plays, rotations, and set patterns.
- Achieved **sub-centimeter tracking precision** on player and ball trajectories by pairing a **YOLOv11** detection model with a **ReID** module to preserve player identity through occlusion, enabling continuous real-time 3D mapping across full match footage.

## Technical Skills

**Languages:** Java, Python, C++, C#, Go, Rust, TypeScript, C, JavaScript, Ruby, Rails, SQL

**Libraries and Frameworks:** Hugging Face, PyTorch, NumPy, Scikit-learn, React, Node.js, FastAPI

**Platforms & Technologies:** AWS, Distributed Systems, Automated Reasoning, AI-Assisted Development, CI/CD, NoSQL (Neo4j), Kubernetes, Docker, Kafka, Jenkins, Terraform, REST APIs, Git, Redis, MCP, RTOS